

A closed-form law for minimum induced drag under power-law bending constraints

Manthan · October 2026 · code: github.com/Manthonix/wing-loading-tradeoff

Problem

Elliptical loading minimises induced drag for a given lift and span, but lift near the tips has a long moment arm about the root, so it is optimal only if structure is free. Here the bending moment is constrained with a general moment arm $|y|^k$: $k = 1$ is root bending, $k = 2$ is Prandtl's integrated bending. Writing circulation as a Fourier sine series, lift depends on A_1 alone, induced drag on $\sum n A_n^2$, and bending on $\sum A_n C_n$: a quadratic objective under linear constraints, solved in closed form by Lagrange multipliers.

Result 1 · the trade-off at fixed span

$$\frac{C_{D,i}}{C_{D,i,\text{ell}}} = 1 + c(k) (1 - f)^2, \quad c(k) = \frac{4(1+k)}{k^2}$$

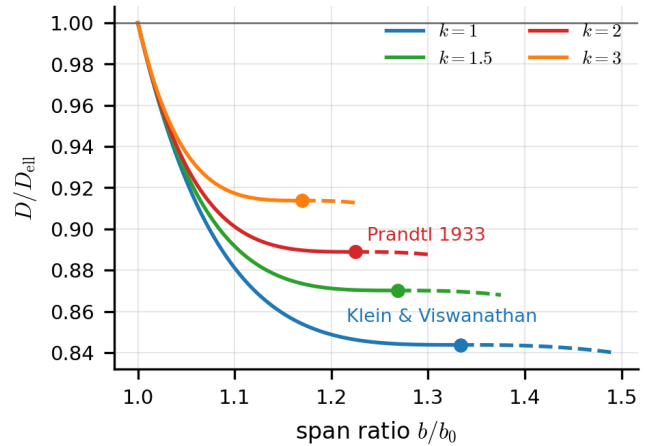
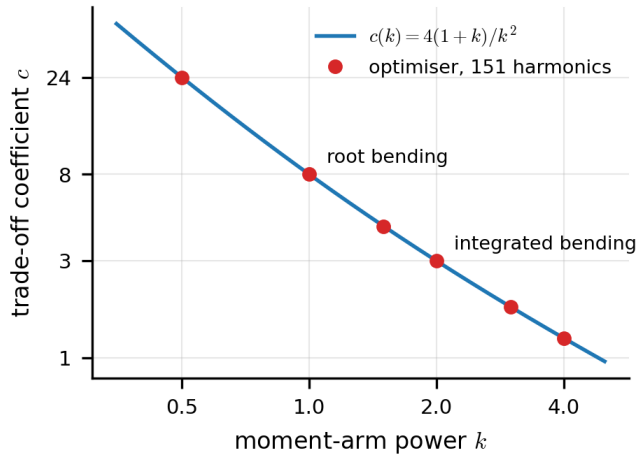
f is bending as a fraction of the elliptical wing's. $k = 1$ gives $c = 8$ and $k = 2$ gives $c = 3$. The penalty is quadratic: small bending relief is nearly free, large relief is expensive.

Result 2 · spending the relief on span

$$\frac{D}{D_{\text{ell}}} = [1 + c(1 - f)^2] f^{2/k}, \quad f^* = \frac{2+k}{2(1+k)}, \quad \frac{b}{b_0} = \left(\frac{2(1+k)}{2+k} \right)^{1/k}$$

Holding lift and bending fixed while span grows, the drag curve's derivative contains a quadratic with discriminant $c[ck^2 - 4(1+k)]$, which vanishes identically for $c = c(k)$. So for every k the only stationary point is a degenerate inflection, not a minimum, and two classical results appear as points on one curve:

k	criterion	f*	span	induced drag	previously known as
1	root bending	3/4	+33.3%	-15.6%	Klein & Viswanathan (1973)
1.5		0.700	+26.8%	-13.0%	
2	integrated bending	2/3	+22.5%	-11.1%	Prandtl (1933)
3		0.625	+17.0%	-8.6%	



Trade-off coefficient: optimiser (markers) against closed form (line).

Drag against span at fixed lift and bending. Dots: inflections. Dashed: tips carry download.

Proof sketch

$$\sum_{n \text{ odd}} \frac{C_n^2}{n C_1^2} = \frac{(1+k/2)^2}{1+k} \implies c = \frac{C_1^2}{\sum_{n \geq 3} C_n^2/n} = \frac{4(1+k)}{k^2}$$

Folding the bending integral onto $[0, \pi/2]$ and applying product-to-sum writes each C_n as a difference of two tabulated integrals of $\cos^k \varphi \cos m\varphi$ (Gradshteyn & Ryzhik 3.631.9), which combine into a single Gamma-function ratio. The sum for c becomes a very-well-poised ${}_5F_4$ series, evaluated by the Rogers–Dougall theorem (DLMF 16.4.9).

Validation and status

Validated: Prandtl's bell recovered exactly ($A_3/A_1 = -1/3$ at $f = 2/3$, $k = 2$); $c(k)$ confirmed three independent ways (optimiser with adaptive quadrature, closed-form coefficients, hypergeometric series) to within 2.5×10^{-7} for $k = 0.5$ to 4 ; numerical derivative of the drag curve never meaningfully negative; result unchanged under $AR \rightarrow \beta AR$, so Mach-invariant below M_{cr} . **Exactness matters:** truncating at seven harmonics gives $c = 8.018$ and produces a spurious minimum at span 1.31. **Observed, not proven:** at f^* the optimal loading meets the tips with zero slope, so f^* is the last point at which the whole wing lifts; this explains why Prandtl's bell sits at $f = 2/3$. **Open:** a physical reason for the form of $c(k)$.

Prior work

The $k = 1$ and $k = 2$ problems are classical: Prandtl (1933), Jones (NACA TN 2249, 1950), Klein and Viswanathan (1973), with related constraints in DeYoung (1979), Pate and German (2013) and Phillips, Hunsaker and Joo (2019), and rigorous treatments of $k = 2$ by Ożański (2024) and Karakhanyan and Katgi (2026). To my knowledge the general coefficient $c(k)$ and the inflection result for every k have not been reported; I would welcome correction. I used AI assistance (Claude) substantially for the derivation and code, and have worked through and reproduced every result.